



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CSA0618 – DESIGN AND ANALYSIS OF ALGORITHM

EXPERIMENT NO: 10

Name: Dharshana Sivakumar

Reg No: 192511054

Program:

```
#include <stdio.h>

int main()
{
    int a[] = {12, 11, 13, 5, 6};
    int n = 5, i, j, key;
    printf("Name: Dharshana Sivakumar\n");
    printf("Insertion Sort\n\n");
    for (i = 1; i < n; i++)
    {
        key = a[i];
        j = i - 1;

        while (j >= 0 && a[j] > key)
        {
            a[j + 1] = a[j];
            j--;
        }

        a[j + 1] = key;
    }
}
```

```

    }

    printf("Sorted Array: [");

    for (i = 0; i < n; i++)
    {
        printf("%d", a[i]);

        if (i < n - 1)
            printf(", ");
    }

    printf("]");

return 0;
}

```

Output:

main.c	Run ▶	Output
<pre> 5 int a[] = {12, 11, 13, 5, 6}; 6 int n = 5, i, j, key; 7 8 printf("Name: Dharshana Sivakumar\n"); 9 printf("Insertion Sort\n\n"); 10 11 for (i = 1; i < n; i++) 12 { 13 key = a[i]; 14 j = i - 1; 15 16 while (j >= 0 && a[j] > key) 17 { 18 a[j + 1] = a[j]; 19 j--; 20 } 21 22 a[j + 1] = key; 23 } 24 25 printf("Sorted Array: ["); 26 for (i = 0; i < n; i++) 27 { 28 printf("%d", a[i]); 29 if (i < n - 1) 30 printf(", "); 31 } 32 printf("]"); 33 34 return 0; 35 } </pre>		<pre> Name: Dharshana Sivakumar Insertion Sort Sorted Array: [5, 6, 11, 12, 13] </pre>